

Code used to get the results

```
# Load required package
```

```
library(vegan)
```

```
# Read the data file
```

```
data <- read.table("data.txt", header = TRUE, sep = "\t", row.names = 1,  
check.names = FALSE)
```

```
# Display data summary
```

```
cat("\n=== DATA IMPORTED ===\n")
```

```
cat(paste("Number of sites:", nrow(data), "\n"))
```

```
cat(paste("Number of species:", ncol(data), "\n"))
```

```
cat("Site names:", paste(rownames(data), collapse = ", "), "\n\n")
```

```
# Save data summary
```

```
sink("1_Data_Summary.txt")
```

```
cat("=====\n")
```

```
cat("DATA SUMMARY\n")
```

```
cat("=====\n\n")
```

```
cat(paste("Number of sites:", nrow(data), "\n"))
```

```
cat(paste("Number of species:", ncol(data), "\n\n"))
```

```
cat("Site names:\n")
```

```
cat(paste(rownames(data), collapse = "\n"))
```

```
cat("\n\nSpecies list:\n")
```

```
cat(paste(colnames(data), collapse = "\n"))
```

```
cat("\n\nAbundance summary per site:\n")
```

```
print(summary(rowSums(data)))
```

```
cat("\n\nSpecies richness per site:\n")
```

```
print(summary(rowSums(data > 0)))
```

```
sink()
```

```
#
```

```
=====
```

```
# 2. NMDS WITH BRAY-CURTIS DISTANCE
```

```
#
```

```
=====
```

```
set.seed(123) # for reproducibility
```

```
nmds <- metaMDS(data, distance = "bray", k = 2, trymax = 100)
```

```
# Save NMDS summary
```

```
sink("2_NMDS_Summary.txt")
```

```
cat("=====\n")
```

```
cat("NMDS RESULTS (Bray-Curtis Distance)\n")
```

```
cat("=====\n\n")
```

```
cat(paste("NMDS stress:", round(nmds$stress, 6), "\n"))
```

```
cat(paste("Number of dimensions:", nmds$ndim, "\n"))
```

```

cat(paste("Number of iterations:", nmds$iterations, "\n"))
cat(paste("Converged:", nmds$converged, "\n\n"))

cat("Stress interpretation:\n")
if(nmds$stress < 0.05) {
  cat(" EXCELLENT: Stress < 0.05\n")
} else if(nmds$stress < 0.1) {
  cat(" GOOD: Stress < 0.1\n")
} else if(nmds$stress < 0.2) {
  cat(" FAIR: Stress < 0.2 (acceptable)\n")
} else {
  cat(" POOR: Stress > 0.2 (caution needed)\n")
}

cat("\nNMDS site scores (coordinates):\n")
nmds_scores <- scores(nmds, display = "sites")
colnames(nmds_scores) <- c("NMDS1", "NMDS2")
print(round(nmds_scores, 4))

sink()

# Save NMDS coordinates as CSV
write.csv(cbind(Site = rownames(nmds_scores), nmds_scores),
  "2_NMDS_Site_Coordinates.csv", row.names = FALSE)

```

```

#
=====

# 3. SIGNIFICANT SPECIES VECTORS (envfit,  $p \leq 0.05$ )
#
=====

set.seed(123)

fit_sp <- envfit(nmds, data, permutations = 999)

# Save all envfit results
sink("3_Envfit_All_Species.txt")
cat("=====\n")
cat("ENVFIT RESULTS - ALL SPECIES\n")
cat("=====\n\n")
print(fit_sp)
sink()

# Extract significant species ( $p \leq 0.05$ )
sig_idx <- which(fit_sp$vectors$pvals <= 0.05)

if(length(sig_idx) > 0) {
  sig_species <- data.frame(
    Species = names(fit_sp$vectors$pvals[sig_idx]),

```

```

    NMDS1 = fit_sp$vectors$arrows[sig_idx, 1],
    NMDS2 = fit_sp$vectors$arrows[sig_idx, 2],
    R_squared = fit_sp$vectors$r[sig_idx],
    P_value = fit_sp$vectors$pvals[sig_idx]
)

sig_species <- sig_species[order(sig_species$R_squared, decreasing = TRUE), ]

# Save significant species as CSV
write.csv(sig_species, "3_Significant_Species_p0.05.csv", row.names = FALSE)

# Save formatted summary
sink("3_Significant_Species_Summary.txt")

cat("=====\n")

cat(paste("SIGNIFICANT SPECIES ( $p \leq 0.05$ ) -", length(sig_idx), "species
found\n"))

cat("=====\n\n")

print(sig_species)

cat("\n\nInterpretation:\n")

cat("These species significantly correlate with the NMDS ordination axes,\n")

cat("indicating they are important drivers of community composition patterns.\n")

sink()
} else {

```

```

writeLines("No significant species found at  $p \leq 0.05$ ",
"3_Significant_Species_Summary.txt")

cat("\nNo significant species found at  $p \leq 0.05$ \n")
}

```

```

#
=====
=====

```

```

# 4. UPGMA CLUSTERING (hclust with method = "average")

```

```

#
=====
=====

```

```

# Calculate Bray-Curtis distance matrix

```

```

bray_dist <- vegdist(data, method = "bray")

```

```

# Perform UPGMA clustering

```

```

upgma <- hclust(bray_dist, method = "average")

```

```

# Save clustering summary

```

```

sink("4_UPGMA_Clustering_Summary.txt")

```

```

cat("=====\n")

```

```

cat("UPGMA CLUSTERING RESULTS\n")

```

```

cat("=====\n\n")

```

```

cat(paste("Clustering method: UPGMA (average linkage)\n"))

```

```

cat(paste("Distance method: Bray-Curtis\n"))

cat(paste("Number of sites clustered:", length(upgma$labels), "\n\n"))

# Cophenetic correlation
cophenetic_cor <- cor(bray_dist, cophenetic(upgma))
cat(paste("Cophenetic correlation coefficient:", round(cophenetic_cor, 4), "\n"))
if(cophenetic_cor > 0.8) {
  cat(" Very good fit (>0.8)\n")
} else if(cophenetic_cor > 0.7) {
  cat(" Good fit (>0.7)\n")
} else {
  cat(" Moderate/poor fit\n")
}
sink()

# Save dendrogram
png("4_UPGMA_Dendrogram.png", width = 1000, height = 700)
par(mar = c(5, 4, 4, 8))
plot(upgma, main = "UPGMA Dendrogram (Bray-Curtis Distance)",
      xlab = "Sites", ylab = "Bray-Curtis Distance", sub = "", cex = 0.8)
rect.hclust(upgma, k = 2, border = "red", lwd = 2)
rect.hclust(upgma, k = 3, border = "blue", lty = 2, lwd = 2)
legend("topright", inset = c(-0.1, 0), legend = c("2 clusters", "3 clusters"),

```

```
col = c("red", "blue"), lty = c(1, 2), lwd = 2, bty = "n", xpd = TRUE)
dev.off()
```

```
# Also save as PDF
```

```
pdf("4_UPGMA_Dendrogram.pdf", width = 10, height = 7)
```

```
par(mar = c(5, 4, 4, 8))
```

```
plot(upgma, main = "UPGMA Dendrogram (Bray-Curtis Distance)",
```

```
      xlab = "Sites", ylab = "Bray-Curtis Distance", sub = "", cex = 0.8)
```

```
rect.hclust(upgma, k = 2, border = "red", lwd = 2)
```

```
rect.hclust(upgma, k = 3, border = "blue", lty = 2, lwd = 2)
```

```
legend("topright", inset = c(-0.1, 0), legend = c("2 clusters", "3 clusters"),
```

```
      col = c("red", "blue"), lty = c(1, 2), lwd = 2, bty = "n", xpd = TRUE)
```

```
dev.off()
```

```
#
```

```
=====
```

```
# 5. CUT DENDROGRAM INTO 2-3 CLUSTERS
```

```
#
```

```
=====
```

```
# Cut for 2 clusters
```

```
clusters_2 <- cutree(upgma, k = 2)
```

```

# Cut for 3 clusters

clusters_3 <- cutree(upgma, k = 3)


# Save cluster assignments

cluster_df <- data.frame(
  Site = rownames(data),
  Cluster_2 = clusters_2,
  Cluster_3 = clusters_3
)

write.csv(cluster_df, "5_Cluster_Assignments.csv", row.names = FALSE)


# Save detailed cluster composition

sink("5_Cluster_Composition.txt")

cat("=====\n")

cat("CLUSTER COMPOSITION\n")

cat("=====\n\n")


cat("FOR 2 CLUSTERS:\n")

cat("-----\n")

for(i in 1:2) {
  sites <- names(clusters_2[clusters_2 == i])

  cat(paste("\nCluster", i, "(", length(sites), "sites):\n"))

  cat(paste(" ", paste(sites, collapse = "\n "), "\n"))
}

```

```
}
```

```
cat("\n\nFOR 3 CLUSTERS:\n")
```

```
cat("-----\n")
```

```
for(i in 1:3) {
```

```
  sites <- names(clusters_3[clusters_3 == i])
```

```
  if(length(sites) > 0) {
```

```
    cat(paste("\nCluster", i, "(", length(sites), "sites):\n"))
```

```
    cat(paste(" ", paste(sites, collapse = "\n "), "\n"))
```

```
  }
```

```
}
```

```
sink()
```

```
#
```

```
=====
```

```
=====
```

```
# 6. NMDS PLOT WITH: Points coloured by cluster, confidence ellipses,
```

```
#   significant species arrows, site labels without overlap (orditorp)
```

```
#
```

```
=====
```

```
=====
```

```
# Create colors for clusters (using 2 clusters for main plot)
```

```
cols_2 <- c("red", "blue")[clusters_2]
```

```
cluster_factor_2 <- factor(clusters_2, labels = c("Cluster 1", "Cluster 2"))

# Main plot with 2 clusters

png("6_NMDS_Plot_2Clusters.png", width = 1100, height = 850)

par(mar = c(5, 5, 4, 2))

# Base plot

plot(nmds, type = "n",

     main = paste("NMDS Ordination of Chrysophyte Communities\n(Stress =",
round(nmds$stress, 4), ")"),

     xlab = "NMDS Axis 1", ylab = "NMDS Axis 2", cex.main = 1.2)

# Add confidence ellipses (ordiellipse)

ordiellipse(nmds, groups = cluster_factor_2,

            col = c("red", "blue"), kind = "sd", conf = 0.95, lwd = 2, label = TRUE)

# Add points coloured by cluster

points(nmds, col = cols_2, pch = 19, cex = 2.5)

# Add site labels without overlap (orditorp)

orditorp(nmds, display = "sites", labels = rownames(data),

         air = 0.3, cex = 0.9, col = "black", pch = NA)
```

```

# Add significant species arrows (plot with p.max=0.05)
if(length(sig_idx) > 0) {
  plot(fit_sp, p.max = 0.05, col = "darkgreen", cex = 1, add = TRUE)
}

# Add legend
legend("topright",
  legend = c("Cluster 1", "Cluster 2", "Significant species ( $p \leq 0.05$ )"),
  col = c("red", "blue", "darkgreen"),
  pch = c(19, 19, NA),
  lty = c(NA, NA, 1),
  lwd = c(NA, NA, 1.5),
  pt.cex = c(1.5, 1.5, NA),
  bty = "n")

grid()
dev.off()

# Also save as PDF for publication quality
pdf("6_NMDS_Plot_2Clusters.pdf", width = 11, height = 8.5)
plot(nmfs, type = "n",
  main = paste("NMDS Ordination of Chrysophyte Communities\n(Stress =",
    round(nmfs$stress, 4), "%)"),

```

```

xlab = "NMDS Axis 1", ylab = "NMDS Axis 2", cex.main = 1.2)
ordiellipse(nmds, groups = cluster_factor_2,
            col = c("red", "blue"), kind = "sd", conf = 0.95, lwd = 2, label = TRUE)
points(nmds, col = cols_2, pch = 19, cex = 2.5)
orditorp(nmds, display = "sites", labels = rownames(data),
        air = 0.3, cex = 0.9, col = "black", pch = NA)
if(length(sig_idx) > 0) {
  plot(fit_sp, p.max = 0.05, col = "darkgreen", cex = 1, add = TRUE)
}
legend("topright",
      legend = c("Cluster 1", "Cluster 2", "Significant species ( $p \leq 0.05$ )"),
      col = c("red", "blue", "darkgreen"),
      pch = c(19, 19, NA),
      lty = c(NA, NA, 1),
      lwd = c(NA, NA, 1.5),
      pt.cex = c(1.5, 1.5, NA),
      bty = "n")
grid()
dev.off()

# Optional: Plot with 3 clusters
cols_3 <- c("red", "blue", "green")[clusters_3]

```

```
cluster_factor_3 <- factor(clusters_3, labels = c("Cluster 1", "Cluster 2", "Cluster 3"))
```

```
png("6_NMDS_Plot_3Clusters.png", width = 1100, height = 850)
```

```
par(mar = c(5, 5, 4, 2))
```

```
plot(nmds, type = "n",
```

```
  main = paste("NMDS Ordination - 3 Clusters\n(Stress =", round(nmds$stress, 4), ")"),
```

```
  xlab = "NMDS Axis 1", ylab = "NMDS Axis 2", cex.main = 1.2)
```

```
ordiellipse(nmds, groups = cluster_factor_3,
```

```
  col = c("red", "blue", "green"), kind = "sd", conf = 0.95, lwd = 2, label = TRUE)
```

```
points(nmds, col = cols_3, pch = 19, cex = 2.5)
```

```
orditorp(nmds, display = "sites", labels = rownames(data),
```

```
  air = 0.3, cex = 0.9, col = "black", pch = NA)
```

```
if(length(sig_idx) > 0) {
```

```
  plot(fit_sp, p.max = 0.05, col = "darkgreen", cex = 1, add = TRUE)
```

```
}
```

```
legend("topright",
```

```
  legend = c("Cluster 1", "Cluster 2", "Cluster 3", "Significant species (p≤0.05)"),
```

```
  col = c("red", "blue", "green", "darkgreen"),
```

```
  pch = c(19, 19, 19, NA),
```

```
  lty = c(NA, NA, NA, 1),
```

```

    pt.cex = c(1.5, 1.5, 1.5, NA),
    bty = "n")

grid()

dev.off()

#
=====

=====

# 7. PERMANOVA (adonis2, Bray-Curtis, 999 permutations)

#
=====

=====

set.seed(456)

# PERMANOVA for 2 clusters

permanova_2 <- adonis2(data ~ clusters_2, method = "bray", permutations = 999)

# PERMANOVA for 3 clusters

permanova_3 <- adonis2(data ~ clusters_3, method = "bray", permutations = 999)

# Extract results for 2 clusters

R2_2 <- permanova_2$R2[1]

p_val_2 <- permanova_2$`Pr(>F)`[1]

```

```

# Extract results for 3 clusters

R2_3 <- permanova_3$R2[1]
p_val_3 <- permanova_3$`Pr(>F)`[1]


# Save PERMANOVA results

sink("7_PERMANOVA_Results.txt")

cat("=====\n")
cat("PERMANOVA RESULTS (Bray-Curtis Distance)\n")
cat("=====\n\n")
cat("Null hypothesis: No difference in species composition between clusters\n")
cat(paste("Permutations:", 999, "\n"))
cat(paste("Distance method: Bray-Curtis\n\n"))


cat("-----\n")
cat("FOR 2 CLUSTERS\n")
cat("-----\n")

print(permanova_2)

cat("\n")

cat(paste("R² =", round(R2_2, 4), "\n"))
cat(paste("R² (%) =", round(R2_2 * 100, 2), "% of variation explained\n"))
cat(paste("p-value =", p_val_2, "\n"))

```

```

if(p_val_2 < 0.05) {
  cat("\n✓ CONCLUSION (2 clusters): Differences are STATISTICALLY
SIGNIFICANT (p < 0.05)\n")

  cat(" The two clusters represent distinct community types.\n")
} else {
  cat("\n✗ CONCLUSION (2 clusters): Differences are NOT significant (p >
0.05)\n")
}

cat("\n\n-----\n")

cat("FOR 3 CLUSTERS\n")

cat("-----\n")

print(permanova_3)

cat("\n")

cat(paste("R² =", round(R2_3, 4), "\n"))

cat(paste("R² (%) =", round(R2_3 * 100, 2), "% of variation explained\n"))

cat(paste("p-value =", p_val_3, "\n"))

if(p_val_3 < 0.05) {
  cat("\n✓ CONCLUSION (3 clusters): Differences are STATISTICALLY
SIGNIFICANT (p < 0.05)\n")

```

```

cat(" The three clusters represent distinct community types.\n")

} else {

  cat("\n ✗ CONCLUSION (3 clusters): Differences are NOT significant (p >
0.05)\n")

}

sink()

#
=====

# 8. PAIRWISE PERMANOVA (if significant for 3 clusters)

#
=====

if(p_val_3 < 0.05) {
  sink("8_Pairwise_PERMANOVA_Results.txt")
  cat("=====\n")
  cat("PAIRWISE PERMANOVA COMPARISONS (3 clusters)\n")
  cat("=====\n\n")

  pairwise_results <- data.frame()

  for(i in 1:2) {

```

```

for(j in (i+1):3) {
  subset_idx <- clusters_3 %in% c(i, j)
  subset_data <- data[subset_idx, ]
  subset_groups <- factor(clusters_3[subset_idx])

  test <- adonis2(subset_data ~ subset_groups, method = "bray", permutations =
999)

  pairwise_results <- rbind(pairwise_results, data.frame(
    Comparison = paste("Cluster", i, "vs Cluster", j),
    F_Statistic = test$F[1],
    R_squared = test$R2[1],
    P_value = test$`Pr(>F)`[1]
  ))
}
}

print(pairwise_results)

cat("\n\nSIGNIFICANT PAIRS ( $p \leq 0.05$ ):\n")
sig_pairs <- pairwise_results[pairwise_results$P_value <= 0.05, ]
if(nrow(sig_pairs) > 0) {
  for(k in 1:nrow(sig_pairs)) {

```

```

        cat(paste(" ✓", sig_pairs$Comparison[k], "(p =", sig_pairs$P_value[k], ")\n"))
    }
} else {
    cat(" No significant pairwise differences detected\n")
}

sink()

write.csv(pairwise_results, "8_Pairwise_PERMANOVA_Results.csv",
row.names = FALSE)
}

#
=====
=====

# 9. COMPREHENSIVE FINAL REPORT

#
=====
=====

sink("9_FINAL_COMPLETE_REPORT.txt")

cat("=====
=====\\n")

cat("TASK 5: CHRYSOPHYTE COMMUNITY ANALYSIS - COMPLETE
REPORT\\n")

```

```
cat("=====  
=====\n\n")
```

```
cat("DATE OF ANALYSIS:", date(), "\n\n")
```

```
cat("-----  
-----  
-----\n")
```

```
cat("1. DATA OVERVIEW\n")
```

```
cat("-----  
-----  
-----\n")
```

```
cat(paste(" • Number of sites:", nrow(data), "\n"))
```

```
cat(paste(" • Number of species:", ncol(data), "\n"))
```

```
cat(paste(" • Total abundance:", sum(data), "\n"))
```

```
cat(paste(" • Mean abundance per site:", round(mean(rowSums(data)), 2), "\n"))
```

```
cat(paste(" • Mean species richness per site:", round(mean(rowSums(data > 0)), 2),  
"\n\n"))
```

```
cat("-----  
-----  
-----\n")
```

```
cat("2. NMDS ORDINATION (Bray-Curtis Distance)\n")
```

```

cat("-----\n")
cat(paste(" • NMDS stress:", round(nmds$stress, 6), "\n"))
if(nmds$stress < 0.05) {
  cat(" • Interpretation: EXCELLENT representation\n")
} else if(nmds$stress < 0.1) {
  cat(" • Interpretation: GOOD representation\n")
} else if(nmds$stress < 0.2) {
  cat(" • Interpretation: FAIR representation (acceptable)\n")
} else {
  cat(" • Interpretation: POOR representation (caution needed)\n")
}
cat(" • Number of dimensions:", nmds$ndim, "\n")
cat(" • Convergence:", nmds$converged, "\n\n")

cat("-----\n")
cat("3. SIGNIFICANT SPECIES (envfit,  $p \leq 0.05$ )\n")
cat("-----\n")
if(length(sig_idx) > 0) {

```

```

cat(paste(" • Number of significant species:", length(sig_idx), "\n"))

cat(" • Top 5 species by R2:\n")

top5 <- head(sig_species, 5)

for(i in 1:nrow(top5)) {

  cat(paste("   -", top5$Species[i], "(R2 =", round(top5$R_squared[i], 4), ", p =",
top5$P_value[i], ")\n"))

}

} else {

  cat(" • No significant species found at  $p \leq 0.05$ \n")

}

cat("\n")

cat("-----\n")

cat("4. UPGMA CLUSTERING (Average Linkage)\n")

cat("-----\n")

cat("-----\n")

cat(paste(" • Cophenetic correlation:", round(cophenetic_cor, 4), "\n"))

cat(paste(" • 2 clusters:", table(clusters_2)[1], "&", table(clusters_2)[2], "sites\n"))

cat(paste(" • 3 clusters:", paste(table(clusters_3), collapse = ", "), "sites\n\n"))

```

```

cat("_____
_____
_____\\n")

cat("5. PERMANOVA RESULTS (Bray-Curtis, 999 permutations)\\n")

cat("_____
_____
_____\\n")

cat("FOR 2 CLUSTERS:\\n")
cat(paste(" • R2 =", round(R2_2, 4), "\\n"))
cat(paste(" • R2 (%) =", round(R2_2 * 100, 2), "%\\n"))
cat(paste(" • p-value =", p_val_2, "\\n"))
if(p_val_2 < 0.05) {
  cat(" • ✓ SIGNIFICANT difference between clusters\\n")
} else {
  cat(" • ✗ NO significant difference between clusters\\n")
}

cat("\\nFOR 3 CLUSTERS:\\n")
cat(paste(" • R2 =", round(R2_3, 4), "\\n"))
cat(paste(" • R2 (%) =", round(R2_3 * 100, 2), "%\\n"))
cat(paste(" • p-value =", p_val_3, "\\n"))
if(p_val_3 < 0.05) {
  cat(" • ✓ SIGNIFICANT difference among clusters\\n")
} else {

```

```

cat(" • ✗ NO significant difference among clusters\n")
}

```

```

cat("\n=====
=====
=====
\n")

```

```

cat("6. FINAL CONCLUSION\n")

```

```

cat("=====
=====
=====
\n\n")

```

```

if(p_val_2 < 0.05) {

```

```

  cat("✓ The clusters identified by UPGMA represent SIGNIFICANTLY
  DIFFERENT\n")

```

```

  cat(" chrysophyte community types based on species composition.\n\n")

```

```

  cat(paste(" Cluster membership explains", round(R2_2 * 100, 2),
  "% of the variation in species composition.\n"))

```

```

} else {

```

```

  cat("✗ No significant differences in species composition were detected\n")

```

```

  cat(" between the clusters identified by UPGMA.\n")

```

```

}

```

```

cat("\n=====
=====
=====
\n")

```